

Total No. of Questions : 8]

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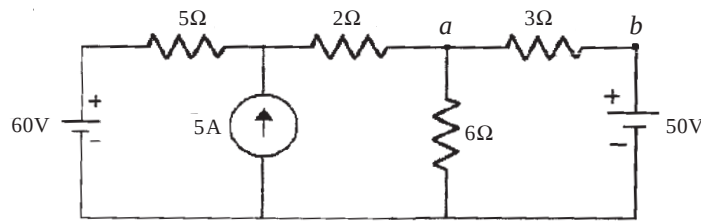
Roll No

EE/EI/IC/EX-305 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Network Analysis
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

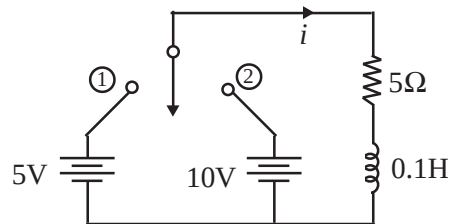
1. Find the current through branch a-b using Mesh analysis shown in figure below.



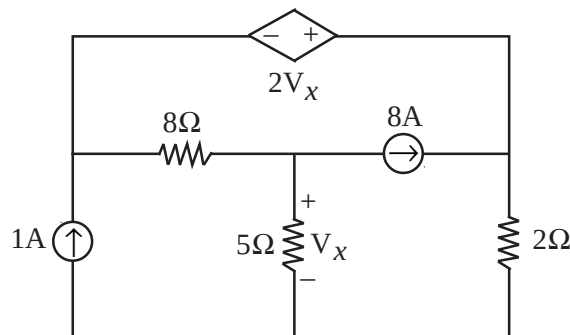
2. Using initial and final value theorems. Determine $f(0)$ and $f(\infty)$ for the following:

i)
$$f(s) = \frac{s^3 + 7s^2 + 5}{s(s^3 + 3s^2 + 4s + 2)}$$

3. In the circuit given below, Obtain the expression for current $i(t)$. When the switch is moved from position (1) to position (2) at $t = 0$.

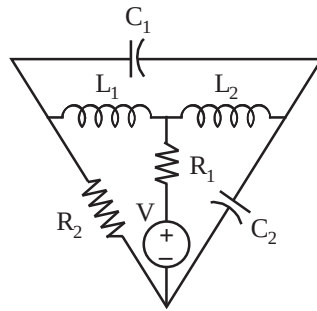


4. Determine the voltage V_x and power supplied by 1A source.



[2]

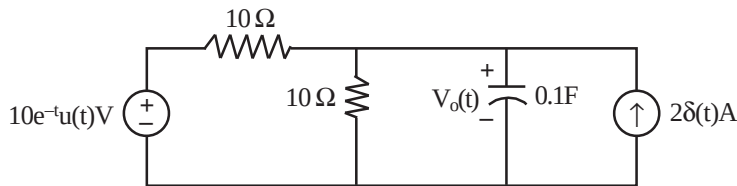
5. Draw the oriented graph and then write the tie-set schedule for the circuit shown in figure.



6. Write short notes on the following:
 i) Dependent and independent sources
 ii) Kirchhoff's voltage and current law
 iii) Dual Network

OR

Find $V_o(t)$ in the circuit of figure, assume $V_o(0) = 5V$.

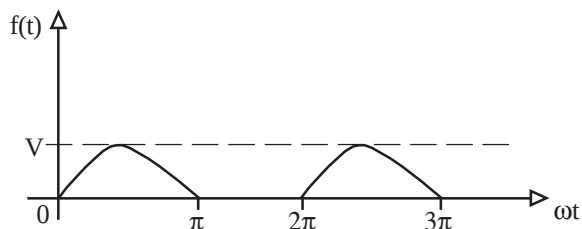


7. Find the exponential Fourier series expansion of the periodic function $f(t) = e^t$, $0 < t < 2\pi$ with $f(t+2\pi) = f(t)$.

OR

Define and distinguish the following network elements with example:

- | | |
|------------------------------------|--------------------------------|
| a) Linear and Nonlinear elements | b) Active and Passive elements |
| c) Lumped and distributed elements | d) Ideal and practical sources |
8. Find the trigonometric Fourier series for the half wave rectified sine wave shown in figure and sketch the amplitude spectrum.



OR

Derive the condition for maximum power transfer in an A.C. circuit with R-L load.
